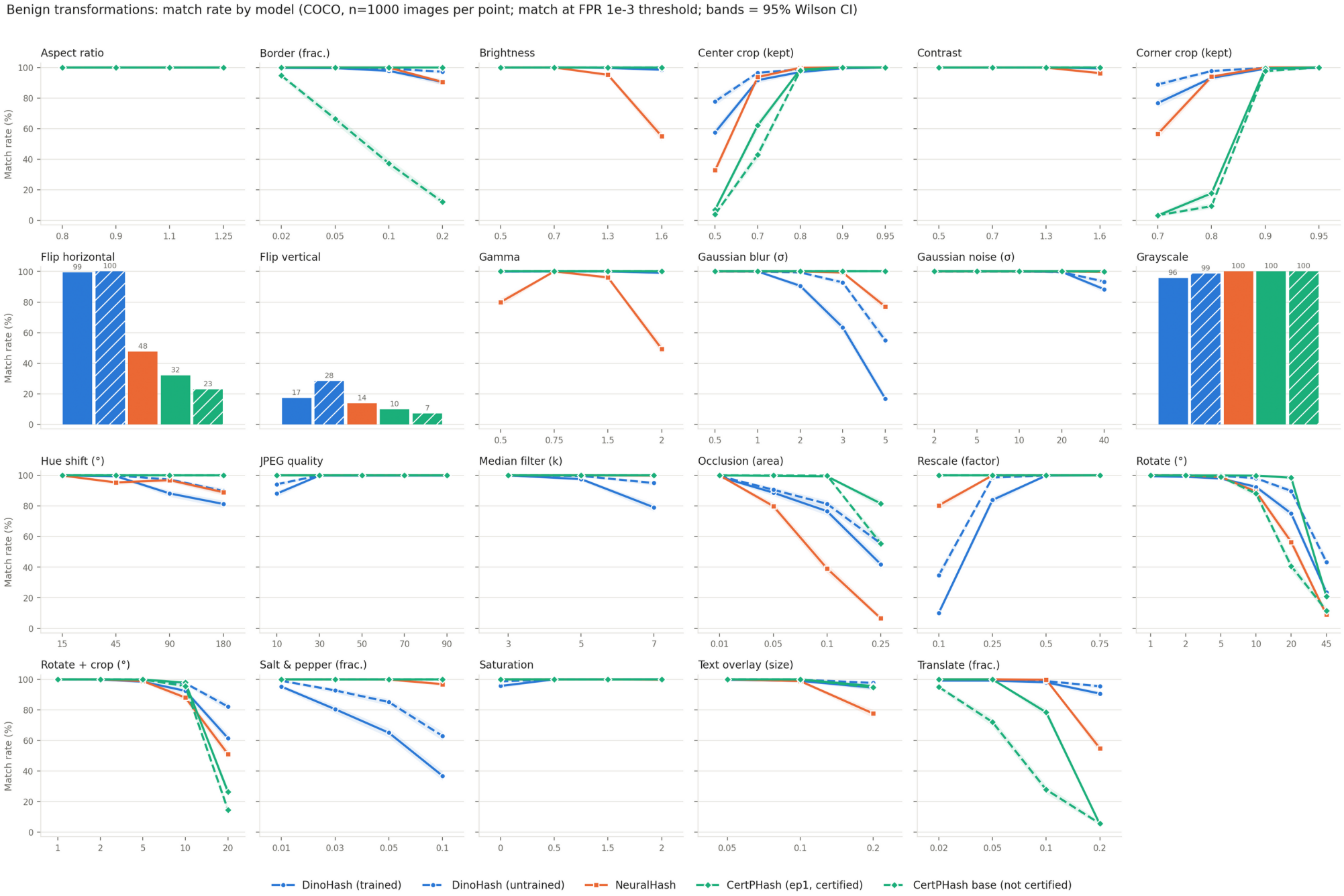
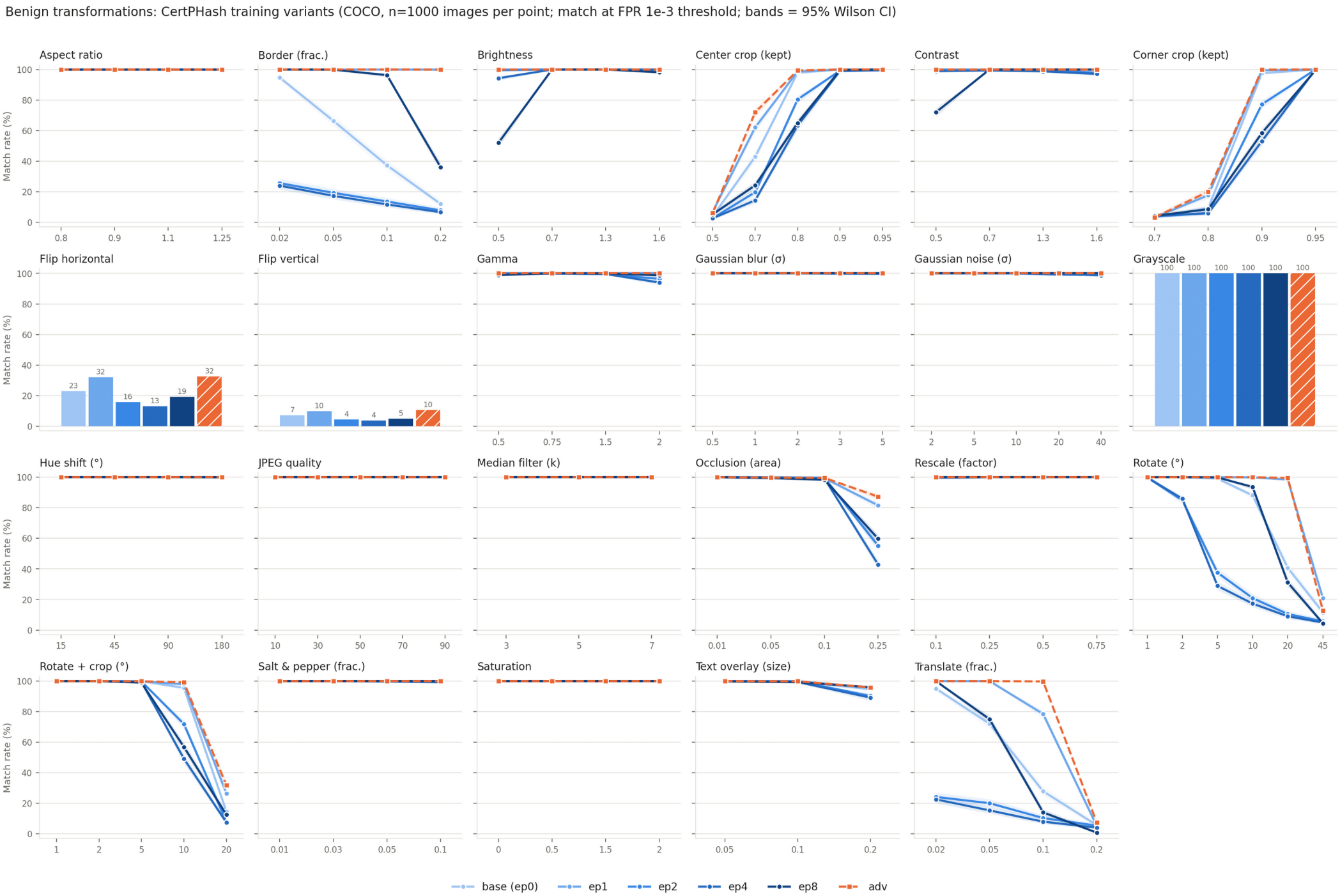
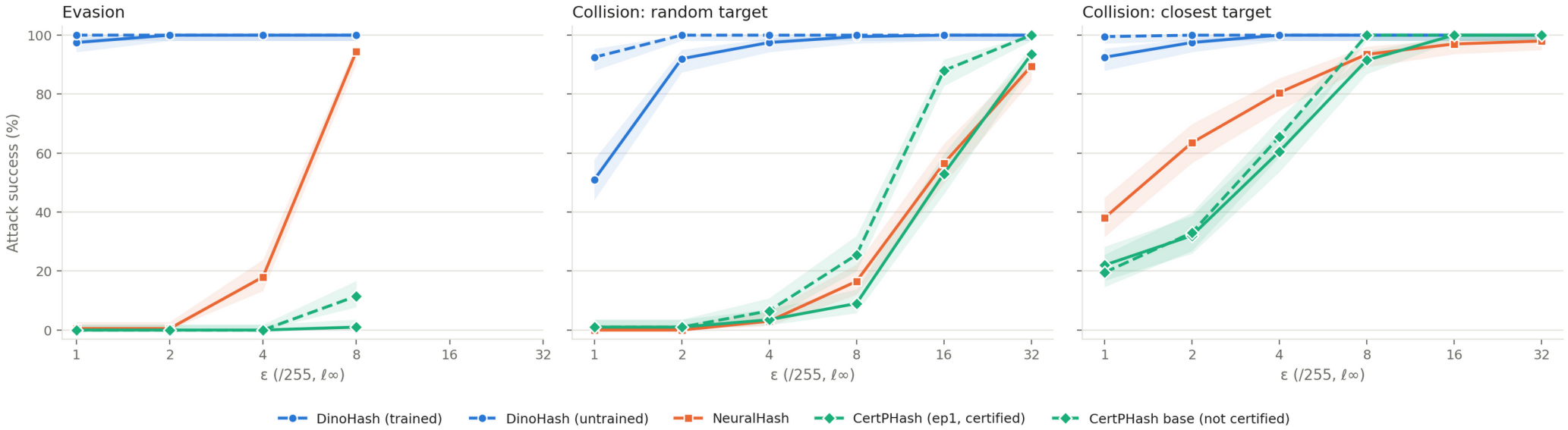
Benign


Benign transformations: match rate by model (COCO, n=1000 images per point; match at FPR 1e-3 threshold; bands = 95% Wilson CI)
Aspect ratio
Border (frac.)
Brightness
Center crop (kept)
Contrast
Corner crop (kept)
Flip horizontal
Flip vertical
Gamma
Gaussian blur (σ)
Gaussian noise (σ)
Grayscale
Hue shift (°)
JPEG quality
Median filter (k)
Occlusion (area)
Rescale (factor)
Rotate (°)
Rotate + crop (°)
Salt & pepper (frac.)
Saturation
Text overlay (size)
Translate (frac.)
Match rate (%)
DinoHash (trained)
DinoHash (untrained)
NeuralHash
CertPHash (ep1, certified)
CertPHash base (not certified)

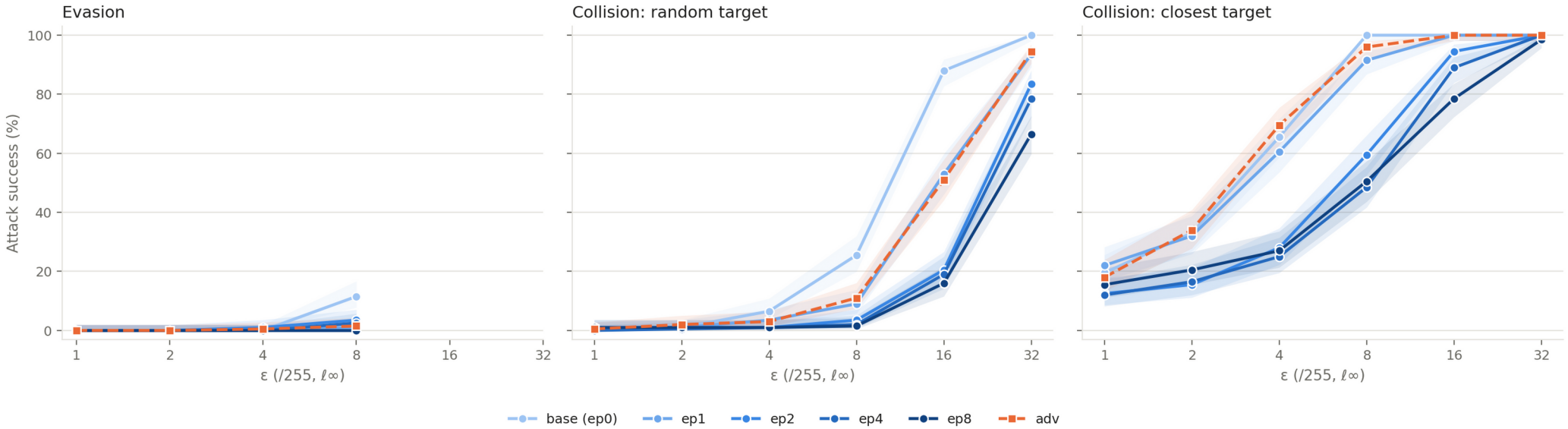

Benign transformations: CertPHash training variants (COCO, n=1000 images per point; match at FPR 1e-3 threshold; bands = 95% Wilson CI)
Aspect ratio
Border (frac.)
Brightness
Center crop (kept)
Contrast
Corner crop (kept)
Flip horizontal
Flip vertical
Gamma
Gaussian blur (σ)
Gaussian noise (σ)
Grayscale
Hue shift (°)
JPEG quality
Median filter (k)
Occlusion (area)
Rescale (factor)
Rotate (°)
Rotate + crop (°)
Salt & pepper (frac.)
Saturation
Text overlay (size)
Translate (frac.)
Match rate (%)
base (ep0)
ep1
ep2
ep4
ep8
adv

Adversarial

White-box attack success by model (V200, n=200 images per point; success at calibrated FPR 1e-3 threshold; bands = 95% Wilson CI)



CertPHash training variants (V200, n=200 images per point; success at calibrated FPR 1e-3 threshold; bands = 95% Wilson CI)



CertPHash variants at the paper's L1 threshold 1800 (V200, n=200; bands = 95% Wilson CI)

